Chapter 3 · Equations and inequalities

1. Simultaneous equations

Solve linear and quadratic systems using elimination or substitution.

2. Equations and graphs

Interpret simultaneous solutions as points of intersection.

3. Solving inequalities

Describe solution sets and combine conditions.

4. Inequalities on graphs

Compare curves and shade regions satisfying several inequalities.

Solutions sets — Luisa · 3

The solutions to an equation or inequality form its **solution set**. We consider real values here.

One value

$2x + 1 = 5$

One value

Several values

$x^2 + 3x + 2 = 0$

Several values

Infinitely many values

$x > 3$

Infinitely many values

No real values

$x^2 = -1$

No real values

Every real value

$x^2 + x = x(x + 1)$

Every real value

Solutions sets — Luisa · 4

A simultaneous solution must satisfy every equation at the same time. Each solution is a pair (x, y).

One solution

$x + y = 9, \quad x - y = 1$

One solution

Two solutions

$x^2 + y^2 = 10, \quad x + y = 4$

Two solutions

No solutions

$x + y = 1, \quad x + y = 3$

No solutions

Infinitely many solutions

$x + y = 1, \quad 2x + 2y = 2$

Infinitely many solutions

1 :: Simultaneous Equations

Luisa · 5

1. Elimination Solve $3x + y = 8$ and $2x - 3y = 9$.

1. Elimination

2. Substitution Solve $x + 2y = 3$ and $x^2 + 3xy = 10$.

2. Substitution

Elimination is convenient for linear equations. For a linear–quadratic system, rearrange the linear equation and substitute into the quadratic. Check each pair in both original equations.

Test Your Understanding

Luisa · 6

Solve simultaneously

$3x^2 + y^2 = 21, \qquad y = x + 1$

Solve simultaneously

4 Solve $x + y = 3$ and $x^2 - 3y = 1$.

4

5. The equations are $y = 2 - 4x$ and $3x^2 + xy + 11 = 0$.

a Eliminate y and show that $x^2 - 2x - 11 = 0$.

a

b Solve the equations, giving exact answers involving $\sqrt{3}$.

b

6. One solution of $y = kx - 5$ and $4x^2 - xy = 6$ is $(1, p)$.

a Find k and p.

a

b Find the other solution.

b

Extension **Luisa · 10**

1. STEP 2010 Q1

Given the identity

$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$

1(a) Find a, b, c, d.

1(a)

1(b) Solve

$5x^2 + 2y^2 - 6xy + 4x - 4y = 9$

$6x^2 + 3y^2 - 8xy + 8x - 8y = 14.$

1(b)

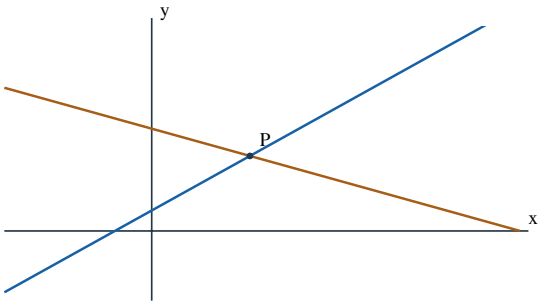
2. MAT 2012 1G For which k do positive real x, y satisfy $2x + ky = 4$, $x + y = k$?

A: all values; B: none; C: only 2; D: only $k > -2$.

2. MAT 2012 1G

Simultaneous Equations and Graphs — Luisa · 11

The graph of an equation contains all points (x, y) satisfying it. For example, $(3, 7)$ lies on $y = 2x + 1$.



Two simultaneous equations

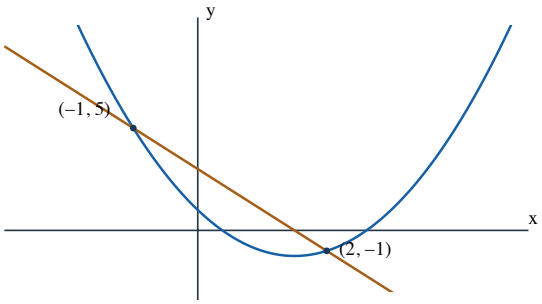
$y = 2x + 1, \qquad x + y = 5$

Which point satisfies both equations?

Two simultaneous equations

Example — Luisa · 12

On the same axes, draw $y = 3 - 2x$ and $y = x^2 - 3x + 1$.



a. Intersections Use the graph to solve the simultaneous equations.

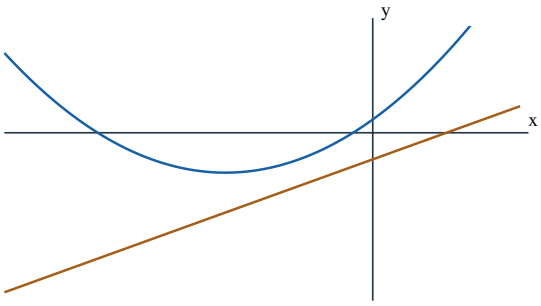
a. Intersections

b. Algebraic check How can we prove that there are two intersections?

b. Algebraic check

Another Example

Draw $y = 2x - 2$ and $y = x^2 + 4x + 1$ on the same axes.



Prove the graphs never meet Solve simultaneously and consider the discriminant.

Prove the graphs never meet

Final Example

The line $y = 2x + 1$ meets $kx^2 + 2y + (k - 2) = 0$ at exactly one point. The constant k is positive.

a Find k.

a

b Find the point of intersection.

b

8. Given $2x - y = 1$, $x^2 + 4ky + 5k = 0$, where $k \neq 0$.

a Show that $x^2 + 8kx + k = 0$.

a

b Given equal roots, find k.

b

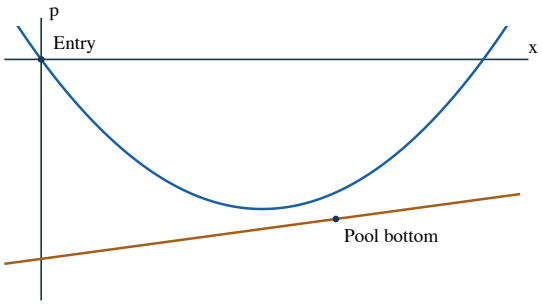
c Find the simultaneous solution.

c

9. A swimmer dives into a pool. Her vertical position p metres is modelled in terms of horizontal distance x metres from entry by

$p = \frac{1}{2}x^2 - 3x$.

The pool bottom is modelled by $p = 0.3x - 6$. Determine whether the swimmer touches the bottom.



9

Set Builder Notation — Luisa · 17

A **set** is a collection of values. Order does not matter and there are no duplicates. Use braces, e.g. $A = \{1, 4, 6, 7\}$.

$A \cap B$: intersection, elements in both sets.

$A \cup B$: union, elements in either set or both.

$\varnothing$: the empty set.

$\mathbb{N}$: positive integers (our convention here).

$\mathbb{Z}$: all integers.

$\mathbb{R}$: all real numbers.

$x \in A$: x is a member of A.

1

$\{1, 2, 3\} \cap \{3, 4, 5\}$

1

2

$\{1, 2, 3\} \cup \{3, 4, 5\}$

2

3

$\{1, 2\} \cap \{3, 4\}$

3

Set Builder Notation — Luisa · 18

Set builder notation describes values without listing them:

{expression : condition}

A colon or vertical bar means “such that”. What do these sets contain?

a

$\{2x : x \in \mathbb{Z}\}$

a

b

$\{2^x : x \in \mathbb{N}\}$

b

c

$\{xy : x, y \text{ are prime}\}$

c

Set builder notation is useful for describing solution sets.

Set Builder Notation — Luisa · 19

Use set builder notation to specify each set. Unless stated otherwise, x is real.

1 All odd integers.

1

2 All real numbers greater than 5.

2

3 All real numbers less than 5 or greater than 7.

3

4 All real numbers between 5 and 7 inclusive.

4

Recap of linear inequalities — Luisa · 20

1

$2x + 1 > 5$

1

2

$3(x - 5) \geq 5 - 2(x - 8)$

2

3

$-x \geq 2$

3

Multiplying or dividing both sides by a negative number reverses the inequality sign.

Combining inequalities If $x < 3$ and $2 \leq x < 4$, what values satisfy both?

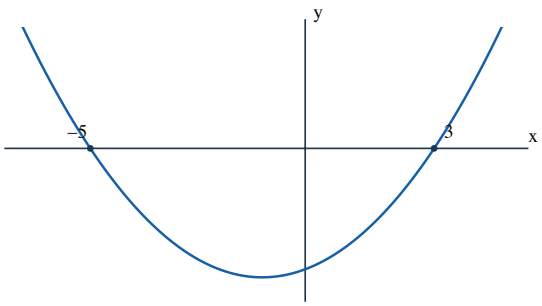
Combining inequalities

RECAP :: Solving Quadratic Inequalities

Luisa · 21

Solve $x^2 + 2x - 15 > 0$.

First put zero on one side, then factorise: $(x + 5)(x - 3) > 0$.



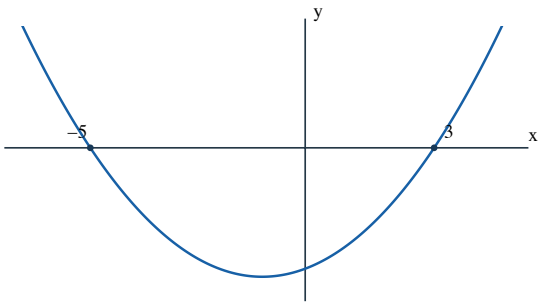
Solution

Solving Quadratic Inequalities

Luisa · 22

Solve $x^2 + 2x - 15 \leq 0$.

Factorise: $(x + 5)(x - 3) \leq 0$.



Solution

Further Examples

Luisa · 23

1

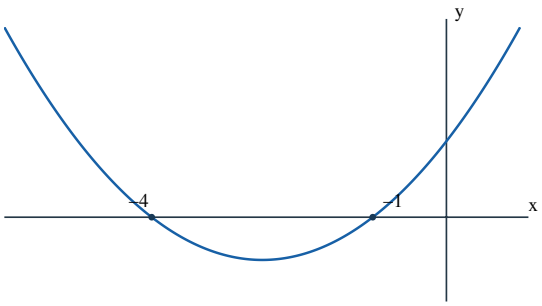
$x^2 + 5x + 4 \geq 0$

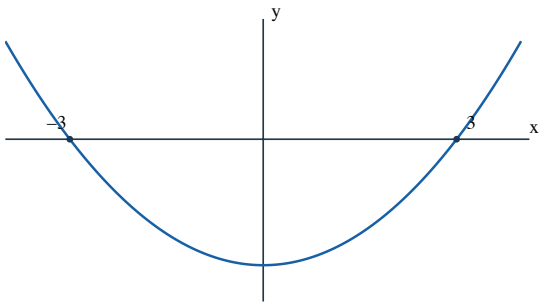
1

2

$x^2 - 9 < 0$

2





Sketch the graph to justify which intervals work. A calculator inequality solver can check your result; choose degree 2.

Test Your Understanding

Luisa · 24

1. Find the set of values of x satisfying each condition.

1(a)

$3(x - 2) < 8 - 2x$

1(a)

1(b)

$(2x - 7)(1 + x) < 0$

1(b)

1(c) Both inequalities above.

1(c)

2. The quadratic equation $2qx^2 + qx - 1 = 0$, with $q \neq 0$, has no real roots.

2(a) Show that $q^2 + 8q < 0$.

2(a)

2(b) Find the possible values of q.

2(b)

We solve an inequality in q to identify equations that have no real solutions in x. These are different variables and different solution sets.

Deal with inequalities with a division by x — Luisa · 25

Find the values satisfying $6/x > 2$, where $x \neq 0$.

Do not simply multiply by x: its unknown sign determines whether the inequality reverses. Instead multiply by $x^2 > 0$.

Working Multiply both sides by x^2, then solve.

Working

Alternatively compare the graphs $y = 6/x$ and $y = 2$. Always keep the original restriction $x \neq 0$.

Exercise 3D/3E – Y1 Page 47-48, 50-51 — Luisa · 26

3(g) Use set notation:

$4(3x + 7) \leq 20$ or $2(3x - 5) \geq \dfrac{7 - 6x}{2}$

3(g)

6 Find the values satisfying $x^2 - 5x - 14 > 0$.

6

8 Given $x \neq 3$, solve $5/(x-3) < 2$.

8

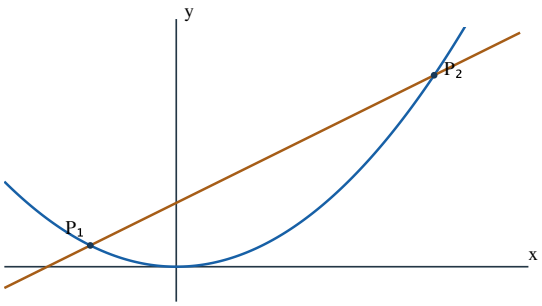
Both endpoints are excluded. At x=3 the original fraction is undefined; at x=11/2 the two sides are equal.

9 The equation $kx^2 - 2kx + 3 = 0$ has no real roots. Prove that $0 \leq k < 3$.

9

Inequalities on Graphs — Luisa · 29

Compare $L_1 : y = 12 + 4x$ and $L_2 : y = x^2$.



a Find their intersections.

a

b Hence solve $12 + 4x > x^2$.

b

Inequality Regions — Luisa · 30

Shade the region satisfying $2y + x < 14$ and $y \geq x^2 - 3x - 4$.

Sketch and shade

1. Draw the boundaries

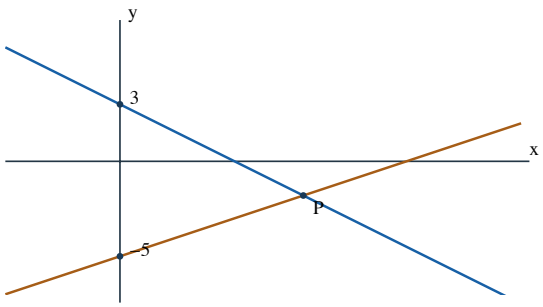
$y = 7 - x/2$: dashed, since the inequality is strict. Its intercepts are $(0, 7)$ and $(14, 0)$.

$y = x^2 - 3x - 4$: solid, since equality is allowed.

Shaded region

Exercise 3F/3G – Y1 Page 53, 55 **Luisa · 31**

1. The lines are $L_1 : 2y + 3x = 6$ and $L_2 : x - y = 5$.



a Find their point of intersection P.

a

b Find where L_1 is above L_2, that is, solve $(6 - 3x)/2 > x - 5$.

b

Exercise 3F/3G – Y1 Page 53, 55 **Luisa · 32**

3. Find where $y = f(x)$ is strictly below $y = g(x)$.

a

$f(x) = 3x^2 - 2x - 1, \quad g(x) = x + 5$

a

b

$f(x) = 2x^2 - 4x + 1, \quad g(x) = 3x - 2$

b

c

$f(x) = 5x - 2x^2 - 4, \quad g(x) = -2x - 1$

c